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USAREUR-AF

Data Architecture Radar -- Plan for Success

Quick Reference

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

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sidebar_position: 3 title: "Plan for Success" description: USAREUR-AF Data Architecture Radar with 5-phase implementation roadmap, approval tracking, and technology standards layer: architecture category: reference

DATA ARCHITECTURE RADAR -- PLAN FOR SUCCESS

USAREUR-AF Enterprise Data Compass and Plan for Success -- Version 1.0 -- February 2026 CUI

Authorizing Document

This radar is the **authoritative record** of data architecture standards, tool selections, and approval status for **USAREUR-AF**.

Role	Signature
Chief Technology Officer, USAREUR-AF	(pending)
Chief Data Architect, ODT	(pending)

BLUF -- Bottom Line Up Front: Definition before data. Doctrine before design. One canonical type per concept. Every object type traces to published doctrine, passes peer review, and meets VAULTIS-A quality standards before it touches production. This plan sequences 5 phases from institutional authority through multi-echelon expansion. **Phase 1 must complete within 30 days** -- without signed authority, everything downstream is advisory. MVP delivery per Secretary of the Army priorities: **speed over perfection, warfighter first.**

ARCHITECTURE RADAR

Ring Definitions

Ring	Description
Adopt	Approved standard. Use it.
Trial	Validated in pilot.

Ring	Description
Assess	Under evaluation.
Hold	Proceed with caution.

5-Layer Data Stack

Layer	Name	Description	Owner
L1	Infrastructure	Compute, network, storage	Platform / DISA
L2	Integration	Pipelines, ingestion, sync	Data Engineers
L3	**Semantic**	**Ontology, meaning, truth** *(Critical)*	**CDA / ODT**
L4	Analytics	BI, models, dashboards	Analysts / ORSA
L5	Activation	Apps, workflows, AI	Product Teams

Investment Rule: L2 + L3 > L5 -- Foundation before features.

Quadrants

Quadrant	Focus
Standards and Governance	Principles, documents, processes
Platform and Tools	Foundry capabilities, tooling
Ontology and Modeling	Object types, patterns, design
Interoperability and Security	CDS, NATO, classification

Key Radar Items by Layer

L1 -- Infrastructure

Item	Ring	Notes
DISA Cloud Hosting	Adopt	FedRAMP-compliant compute, storage, network
Foundry/Vantage Platform Designation	Trial	Formal command guidance pending CTO signature
MSS/Vantage NIPR Instance	Adopt	Primary unclassified operational data environment
Vantage SIPR Instance	Trial	Required for cross-domain architecture
Vantage JWICS Instance	Assess	Under evaluation for USAREUR-AF deployment

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Item	Ring	Notes
Platform Team Ownership Model	Adopt	Clear RACI for compute, network, storage provisioning

L2 -- Integration

Item	Ring	Notes
Pipeline Builder	Adopt	Primary L2 integration tool
Code Workbook	Adopt	Custom data transformations
Data Connections	Adopt	Source system integration
Cross-Domain From Day One	Adopt	CDS topology as first-class design constraint
CDS Export/Import Pattern	Trial	Source -> export filter -> CDS guard -> target -> ontology merge
Foundry Cross-Enclave Sync	Assess	NCDSMO baseline patterns; hardest technical risk
NCDSMO Baseline Compliance	Trial	Cross-domain data flow standards
CDS Latency Documentation	Trial	Per Compass Section 6

L3 -- Semantic (Critical Layer)

Item	Ring	Notes
Definition Before Data	Adopt	Principle: definition grounded in published doctrine
One Canonical Type Per Concept	Adopt	Type proliferation is the primary anti-pattern
Semantic Layer Is Critical Layer	Adopt	L2 + L3 investment takes priority over L5
Governance By Design	Adopt	Peer review, DDOF gates, VAULTIS-A
Object Type Cookbook v2.0	Adopt	9 varieties, decision trees, quality metrics
Cookbook Addendum A	Adopt	Data scope assessment, doctrine alignment
Peer Review (Mandatory)	Adopt	Checklist-driven, reviewer cannot be author
Architecture Review Board	Trial	Biweekly cadence, approve/revise/reject authority
Entity Object Type	Adopt	People, units, equipment
Event Object Type	Adopt	SIGACTs, movements
Capability Object Type	Adopt	Warfighting functions, operational capabilities
Controlled Vocabulary	Adopt	Enumerations and codes
Reference Object (4-Tier)	Adopt	Standardized lookups

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Item	Ring	Notes
Relationship Link	Adopt	Object-to-object connections
Assessment Object Type	Trial	Scores and readiness assessments
Document Object Type	Trial	OPORDs, publications
Metric Object Type	Assess	KPIs and measures
Doctrine-to-Ontology Alignment Worksheet	Adopt	REUSE/EXTEND/CREATE decision framework
Core Ontology Starter Set	Assess	Minimum shared starting point
Bitemporal Modeling	Assess	Valid-time and transaction-time tracking
Classification Ceiling Property	Adopt	Mandatory per Compass Section 6
NAFv4 Compatibility	Assess	Coalition interoperability requirement
DODAF DIV-1/2/3 Alignment	Trial	Architecture viewpoints as ontology artifacts
STANAG Alignment (4559/5500/APP-6)	Assess	NATO interoperability

L4 -- Analytics

Item	Ring	Notes
Contour	Adopt	Interactive data analysis and BI
Workshop Analytics	Trial	Ontology-backed analytics applications
Quiver	Assess	Advanced analytical notebooks
VAULTIS-A Quality Framework	Adopt	8 quality dimensions, 85% minimum gate score
Ontology Health Dashboard	Assess	Object type counts, quality scores, freshness

L5 -- Activation

Item	Ring	Notes
Workshop Apps	Trial	Ontology-backed operational applications
Slate Dashboards	Hold	Legacy; being superseded by Workshop/OSDK
OSDK Front-End	Trial	Custom web apps against Foundry ontology
DDOF Playbook v2.2	Adopt	6-phase data product lifecycle
NAF Architecture Builder	Assess	React graph editor, prototype stage
Course Portal	Adopt	Training slide-viewer with 4 decks

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Item	Ring	Notes
Investment Rule: L2+L3 > L5	Adopt	Foundation before features
Semantic Layer Instructions	Adopt	Doctrine-first governance philosophy

ASSESSMENT OF CURRENT STATE

Strengths

Asset	Status	Assessment
Object Type Cookbook v2 + Addendum A	Complete	Production-grade. 9 object type varieties, decision trees, quality metrics, peer review checklists.
Semantic Layer Instructions	Complete	Doctrinally sound. Definition-first, PR-based governance, full-time stewardship mandate.
Data Modeling Foundations (L1 + L2)	Complete	Strong training curriculum. Covers Kimball, Inmon, Data Vault with Foundry mapping.
2026 Data Stack Deep Dive	Complete	5-layer architecture with Foundry mapping. L3 correctly identified as critical layer.
EA Curriculum (6-part series)	Complete	Comprehensive. Covers foundations through military application.
Tool Convergence Assessment	Complete	Ready for CTO signature. 7 testable hypotheses.
DDOF Playbook v2.2	Complete	6-phase lifecycle with VAULTIS-A quality framework.
Cross-Domain Solutions Reference	Complete	CDS taxonomy, NCDSMO baseline, Foundry cross-enclave sync patterns.
HERMES Program Identity	Complete	Landing page, OPORD 0015-26 context, team roster, timeline.
NAF Architecture Builder	Prototype	React graph editor with xyflow. IR extraction pipeline works.
Course Portal	Complete	Static site serving 4 training decks with slide preview and downloads.
EA vs DA Reference Page	Complete	Clean interactive reference establishing DA as first-class child of EA.

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Gaps

Gap	Risk If Not Addressed
Enterprise Data Compass (signed)	Without the signed Compass, every standard in the Cookbook is advisory, not authoritative.
Starter Ontology (core object types)	Every team builds from scratch. No shared starting point.
Doctrine-to-Ontology Mappings	No traceable thread from DoD Data Strategy to Foundry object type.
Ontology Health Metrics Dashboard	Cannot answer: is the architecture being used?
90-Day Tool Audit Results	The 7 hypotheses remain untested. Leadership cannot act on assertions.
Training Certification Mechanism	Training becomes optional in practice.
Cross-Domain Implementation Guide	Each team figures out CDS independently. Inconsistent patterns.
Sequenced Program Roadmap	No visible critical path. Leadership cannot track progress.

5-PHASE IMPLEMENTATION ROADMAP

Phase 1: Establish Authority (30 days)

Objective: Convert the existing body of work into signed, enforceable institutional authority. Without this, everything downstream is optional.

Deliverables:

1. **Enterprise Data Compass v1.0** -- Compile the Cookbook, Semantic Layer Instructions, and 5-Layer Stack into a single signed reference. CTO and CDAO signature blocks. Version-controlled.
2. **Tool Convergence Directive** -- Get the existing convergence memo signed by the CTO. Issue moratorium on new non-strategic tool adoption.
3. **Foundry/Vantage Platform Designation** -- Formal command guidance designating Foundry/Vantage as the authoritative operational data platform.
4. **HERMES Program Charter** -- Formalize HERMES with scope, authority, resourcing, and reporting chain.

Exit Criteria:

- Enterprise Data Compass signed by CTO/CDAO

- Tool convergence memo signed and moratorium in effect
- Platform designation guidance issued
- HERMES charter approved

Phase 2: Build the Foundation (60 days)

Objective: Deploy the first production ontology objects and complete the first doctrine-to-ontology mapping. Prove the methodology works end-to-end.

Deliverables:

1. **Core Ontology Starter Set** -- Deploy foundational object types: Unit, Person, Equipment (entities) + WarfightingFunction, OperationalDomain, Echelon, ClassificationLevel (CVs) + Capability + SIGACT (event). Each passes Cookbook peer review.
2. **First Doctrine-to-Ontology Mapping** -- Complete Addendum A worksheet for one priority domain (Readiness or Force Structure). Document REUSE/EXTEND/CREATE decisions with full traceability.
3. **CDS Architecture Pattern Validation** -- Implement and validate Foundry cross-enclave sync for at least one object type across NIPR/SIPR.
4. **DDOF Pilot Product (Phase 1-3)** -- Run DDOF playbook end-to-end on one data product. Validate VAULTIS-A quality gates. Document lessons learned.

Exit Criteria:

- Core starter ontology deployed in production Foundry
- At least one doctrine mapping completed with peer review
- CDS sync validated for at least one object type
- One DDOF product through Gate 3 with VAULTIS-A scores documented

Phase 3: Scale Through Training (45 days)

Objective: Ensure every person building on the ontology has completed training and can pass a competency check.

Deliverables:

1. **Training Certification Program** -- Build assessments for each level (Foundations, Level 2, Semantic Layer). Minimum passing score. Certificates tracked in HERMES.
2. **Level 3 Training: Advanced Patterns** -- New deck covering CDS ontology patterns, bitemporal modeling, entity resolution, and DDOF integration.
3. **Leadership Briefing Package** -- Condensed training for G3/G8/CTO audiences. Why it matters, how to consume it, how to govern it.
4. **Train-the-Trainer Cadre** -- Certify 3-5 instructors who can deliver training at echelon without ODT presence.

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Exit Criteria:

- Assessment mechanism deployed for all training levels
- Level 3 deck complete and peer-reviewed
- At least one leadership briefing delivered to GO/SES audience
- 3+ certified trainers outside ODT core team

Phase 4: Govern and Measure (60 days)

Objective: Prove the architecture is being adopted, deliver audit data, and stand up self-sustaining governance.

Deliverables:

1. **90-Day Tool and License Audit** -- Quantify: total license costs by tool, FTEs on platform admin vs. operational output, cross-domain availability, capability overlap with Foundry.
2. **Ontology Health Dashboard** -- Foundry Workshop dashboard: object type count, link coverage, quality scores, consumer count, freshness, VAULTIS-A scores.
3. **Architecture Review Board (ARB)** -- Stand up ARB per EA_04 Governance. Connect to Cookbook peer review and DDOF Phase 1 gate. Charter, criteria, cadence.
4. **Talent Reclamation Report** -- Using audit data, quantify personnel recovered from redundant platform management. Map to Foundry roles.

Exit Criteria:

- Audit report delivered to CTO with quantified findings
- Ontology health dashboard live in production Foundry
- ARB chartered and operational with first review completed
- Talent reclamation report with specific transition recommendations

Phase 5: Expand and Sustain (90+ days)

Objective: Scale beyond USAREUR-AF core to multi-echelon, joint, and coalition adoption.

Deliverables:

1. **Multi-Echelon Deployment** -- Push ontology access and training to division and brigade level. Measure time-to-first-product at each echelon.
2. **Joint Adoption (NAVEUR / USAFE)** -- Execute onboarding plan for NAVEUR and USAFE. Validate semantic layer supports joint operational concepts without forking.
3. **NATO Interoperability Validation** -- Test ontology objects against NAFv4 viewpoints using Architecture Builder. Validate STANAG alignment. Document gaps.

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4. **Program Roadmap (Live Artifact)** -- Deploy AG Grid roadmap with actual program data. Capability gaps mapped to funded initiatives. Monthly update cadence.
5. **Enterprise Data Compass v2.0** -- Incorporate lessons learned from Phases 1-4. Update standards based on production experience. Add joint/coalition patterns. Re-sign.

Exit Criteria:

- At least 2 echelons below theater producing data products on shared ontology
- At least 1 joint component onboarded to shared semantic layer
- NAF alignment assessment completed with documented results
- Live roadmap accessible to command leadership

CRITICAL DEPENDENCIES

Dependency	Blocks	Mitigation
CTO/CDAO signature on Enterprise Data Compass	All of Phase 2-5 (standards become enforceable)	Pre-brief CTO. Use convergence memo as forcing function.
Core ontology starter set deployed	Phase 3 training (need real objects to train against)	Begin design in parallel with Phase 1 signatures.
90-Day audit data	Phase 4 ARB decisions and Phase 5 resource reallocation	Start audit data collection in Phase 1.
Certified trainers	Phase 5 multi-echelon deployment	Identify candidates in Phase 2. Begin certification in Phase 3.
CDS pattern validation	Phase 5 joint and coalition adoption	Start CDS work in Phase 2 and do not stop. Hardest technical risk.

BOTTOM LINE

1. Get the Enterprise Data Compass signed. Without this, the Cookbook is a suggestion. With it, the Cookbook is law.
2. Get the convergence memo signed. The moratorium creates urgency.
3. Deploy the core ontology starter set. Every day without shared objects is a day teams build in isolation.

The doctrine is written. The tools are built. The training exists. Now make it real.

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EXTERNAL ALIGNMENT — ARMY INSTITUTIONAL PROGRAMS

The USAREUR-AF data architecture and training investment aligns with — and in several areas leads — emerging Army institutional programs. Awareness of these validates our approach and identifies future integration opportunities.

Program	Proponent	Status	Alignment
CAC Maven C2 Integration	MCCoE/CGSC/ CAC C2DAO	Active (Feb 2026 T3 session)	Institutional Army integrating Maven into PME and operator training. Our TM-10–30 courses cover the same competencies the 8-hour MCCoE operator course and CAC Builders Course will deliver.
AKMP Data Immersion Course	MCCoE/C2ID	Operational (2nd iteration Feb 2025)	32-hour KM course on Vantage, data governance, GenAI. Aligns to TM-40K.
TRADOC Decision Optimization Branch	TRADOC C2DAO/OCKO	FOC Dec 2025	6–8 simultaneous data/KM training projects. Our Phase 3 (Scale Through Training) parallels TRADOC's timeline.
CALL 25-10, Commander and Staff Guide to Data Literacy	CALL	Published Apr 2025	Institutional recognition that data literacy requires formal education — same premise as our program.
Signal School Data for Leaders MTT	Signal School (Fort Eisenhower)	Operational (Nov 2024 first delivery)	4-day Power BI course for leaders. Complementary to TM-EXEC.
Army Reserve ODT Pilot (Mojave Falcon)	OCAR	Complete (Jun 2025)	Validated that data literacy baseline is prerequisite for ODT effectiveness.
First Army ODT Mobilization Simulation	First Army + Georgia Tech	Operational (2026)	Most advanced publicly documented ODT product. Campaign-level ORSA on Army Data Platform.

Operational Data Team -- HERMES Program -- USAREUR-AF CUI Distribution: USAREUR-AF / LANDCOM -- Operational Data Team, HERMES Program. Authorized recipients only per CUI handling procedures. Proponent: Chief Data Architect, Operational Data Team (ODT) Authority: OPORD 0015-26, USAREUR-AF Unified Data Transformation Strategy

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